Akhil Chatla

System Engineer

A seasoned system engineer with experience in data engineering and frontend development who is recognized for strong technical expertise and teamwork at TCS. Currently pursuing a Master's in Data Science and Analytics at Florida Atlantic University, expected graduation in May 2025. Proficient in data visualization, predictive analytics, and database technologies, with a proven track record of leading impactful projects.

CONTACT

(+1) 561-332-7608

akhilchatla63@gmail.com

EDUCATION

Masters in Data Science and Analytics from Florida Atlantic University, USA

RELEVANT COURSES

- Introduction to Data Science Analytics
- Data Mining & Machine Learning
- Data Mining Predictive Analytics
- Database Management Systems
- Introduction to Business Analytics, Big Data
- Natural Language Processesing

TECHNICAL SKILLS

- **Programming Languages:** PL/SQL, SQL, Python, JavaScript, html, and CSS
- **Data Bases:** MySQL, SQL Server, Oracle.
- **NoSQL Data Bases:** Cassan-dra, HBase, Dynamo DB.
- **Visualization & ETL tools:** Tableau, Power BI, Informatica PowerCenter
- **Cloud Technologies:** Azure & AWS, GCP.
- **IDE's:** Eclipse, Jupyter Notebook, PyCharm
- **Version Control Systems:** Git, Jira, Jenkins
- **Operating Systems and IDE:** Windows, Linux, Unix

  Jupiter, Visual Studio, IDLE

PROFILE SUMMARY

- Around 2.4 years of professional experience in the complete software development lifecycle, encompassing analysis, design, **development**, **testing**, **documentation**, implementation, and maintenance of application software in web-based and client/server environments.
- Proficiency in **Informatica PowerCenter** for designing and implementing robust **ETL** processes, ensuring seamless data integration and transformation.
- Employed the **Agile** paradigm throughout the entire software development life cycle.
- Ability to work effectively and efficiently as a **team member** and **individually** with a desire to learn new skills and technology.
- Work **collaboratively** with cross-functional teams in **agile environments** to align **database** solutions with **business goals**, supporting informed, **data-driven decision-making.**
- Expertise in **PL/SQL development,** optimizing and managing **Oracle databases** for efficient **data** retrieval, processing, and reporting**.**
- Experienced in **Oracle Data Integrator** for designing and optimizing data integration workflows, and **ETL** processes to ensure efficient data flow across diverse systems.
- Developed and implemented **Extract, Transform, and Load** processes to transfer data from **production** systems to the **data warehouse**.
- Proficient in using **PuTTY** to manage and configure **Unix/Linux servers**, with expertise in **KornShell (ksh)** command execution.
- Skilled **in XLMiner** within **MS Excel** for data mining tasks such as predictive modeling, clustering, regression, and time series analysis to extract actionable insights.
- Skilled in **Python** libraries like **NumPy, Pandas, Matplotlib, and TensorFlow** for data analysis, visualization, and building machine learning models to deliver data-driven solutions.
- Proficient in cloud infrastructure management, compute services, storage solutions, and basic networking using **Google Cloud Platform (GCP).**
- Proficient in transforming complex datasets into actionable insights through **Power BI,** with data modeling, **SQL, DAX, and data integration** expertise to deliver impactful and efficient BI solutions.
- Skilled in utilizing **Visual Code** to design and code **responsive front-end interfaces**, ensuring compatibility and interactivity across various devices and aligning with best practices for clean, maintainable **code** using **HTML**, **CSS**, and **Javascript**.

PROJECTS

1).

Predictive Machine Status Analysis Using Sensor Data

Developed a Random Forest model achieving **92.5% accuracy** to predict machine status, validated via **5-fold cross-validation (91.7% mean accuracy**). Preprocessed 15+ sensor readings, identified key correlations (e.g., sensor_03 with 0.87 correlation), and highlighted sensor_05 as the top feature **(24%).** Delivered actionable insights for proactive maintenance to reduce downtime.

Tools: Python, scikit-learn, Pandas, Matplotlib, Random Forest Classifier.

2) Project Highlights: Advanced Text and Audio Classification

Built a BiLSTM-based text classifier with **77.94% accuracy** and **70.45% F1-score** on **10,000+ abstracts**, and an **LSTM-based gunshot classifier** achieving **89.16% accuracy** on **four firearm types**.Compared **LSTM and CNN models** for audio classification, processing **1,500+ samples**, with LSTM outperforming CNN (**77.71% accuracy**).Applied advanced **EDA, MFCC feature extraction**, and evaluation metrics (e.g., precision, recall) to ensure meaningful insights and robust performance.

Tools: Python (TensorFlow, Librosa, Pandas), Jupyter Notebook, Google Colab.

3) Predicting Credit Card Payment Defaults

Developed a predictive model using the UCI Credit Card Dataset (30,000 records) to identify high-risk clients. Preprocessed data with **XLMiner**, applied **PCA** to select **top 5 predictors**, and built models using **KNN** and **Logistic Regression**, achieving the highest accuracy with KNN (Model 5). Delivered actionable insights for proactive risk management and financial interventions.

Tools Used: XLMiner, PCA, KNN, Logistic Regression.

4) Spotify Data Analysis Project (Power BI & Tableau)

Analyzed a dataset of **32,833 Spotify** tracks spanning 6 playlist genres, extracted using the **Spotify API (spotifyr)** and processed with Python **and leveraged** both **Power BI and Tableau** to design 6 interactive dashboards (3 in each tool), providing insights into tracks, albums, playlists, and artists. Key findings included a **20%** rise in album releases, a bell-shaped distribution for track durations **(~200 seconds),** and a uniform distribution of genres and subgenres. Highlighted top 10 artists with average **popularity >40** and growing contributions from independent creators. Used **DAX measures, Power Query**, and Tableau's calculated fields to build **dynamic, drill-down visualizations**. Delivered actionable insights to enhance playlist diversity, optimize energy-tempo relationships, and drive better audience engagement.

Tools: Spotify API, Python (spotifyr and Pandas) for data extraction and preprocessing, and Power BI & Tableau for visualization.

WORK EXPERIENCE

Tata Consultancy Services | HYDERABAD | INDIA

Client: State Bank Global IT Center | Role: Frontend Developer | May 2023-Aug 2023

Description: The State Bank Global IT Center in India is the State Bank of India's central hub for managing its core banking, digital services, IT security, and technological innovations globally.

Responsibilities:

- ❖ Created interactive **Power BI dashboards** that allow users to analyze and **visualize** large **datasets**, making drawing actionable insights easy.
- ❖ Designed **dashboards** with user-friendly features, enabling users to manually **input data** through the interface, store it securely, and retrieve or edit it whenever needed.
- ❖ Integrated data from multiple sources like databases, **APIs**, and files into Power BI, ensuring accurate and seamless reporting.
- ❖ Added features to validate **user-input** data, ensuring consistency and **reliability** while storing and updating information.
- ❖ Enhanced dashboard interactivity with advanced **DAX** measures and options like slicers, filters, and editable fields to improve usability.

- Automated workflows to streamline the process of importing and **transforming data**, reducing manual effort and speeding up updates.
- Optimized dashboards to handle **large datasets** efficiently, ensuring quick load times and a smooth user experience.
- Developed responsive web components using **HTML5, CSS3, and JavaScript** to complement the **dashboards** and provide an engaging experience on any device.
- Used **JavaScript** and **jQuery** to make dashboards more dynamic, allowing users to edit and view custom inputs in real time.
- Collaborated with teams to design **ETL** processes that ensure data is always prepared and ready for analysis.
- Built functionality in **dashboards** to display real-time data, so users always have access to the latest information.
- Maintained a clean, modular codebase **using Visual Studio** Code to ensure ease of **debugging** and future **development**.
- Documented clear user guides and instructions for dashboards, enabling users to input, **update**, and **retrieve** data with minimal technical support.
- Worked closely with stakeholders to understand their data needs and built customized solutions to improve decision-making processes.
- Prioritized **data security** by setting up role-based access controls, ensuring sensitive information was protected while allowing flexibility for users.

Environment: HTML, CSS, Javascript, Oracle, SQL, Power BI, jQuery.

Client: British Telecom | Role: Data Engineer | June 2021 – April 2022

Description: The British Telecom (BT) is a global leader in telecommunications, providing broadband, mobile, and IT services to businesses and governments worldwide.

Responsibilities:

- Designed and implemented **ETL** processes to **load** and **transform** data from the **Data Staging Area (DSA)** to the **Operational Data Warehouse (ODW)** in **Oracle Database**, ensuring seamless data flow.
- **Developed Robust PL/SQL Solutions:** Designed and implemented **PL/SQL procedures** and **packages** to automate workflows for seamless data processing across the **DSA, ODW**, and metadata management in the **Data Repository Area (DRA).**
- Handled large datasets received from upstream systems, ensuring accurate ingestion and maintaining the quality of data throughout the pipeline.
- Utilized **Oracle Data Integrator (ODI)** to streamline data integration and transformation processes, following the ELT methodology for improved performance.
- **Streamlined Data Integration with Scripting:** Utilized **KSH scripts** on **Putty** servers to execute data loading jobs, ensuring accurate population of data into specific schemas and enabling efficient **ETL workflows**.
- **Optimized Workflow Execution:** Integrated PL/SQL procedures with **KSH scripts** to coordinate the loading of transformed data from the staging area to target schemas, ensuring smooth and reliable pipeline operations.
- Automated data transfer between layers in the **Oracle database** by running **KornShell (KSH)** scripts on Putty servers, enabling seamless data loading and transformation from **DSA to ODW**.
- **Optimized Data Integration Workflows:** Identified and resolved data loading issues during ETL processes in **Oracle Data Integrator** (ODI) by analyzing session logs, validating transformation mappings, and debugging Knowledge Modules (KMs).
- **Cross-Schema Data Management:** Successfully managed data transfers and transformations between schemas by monitoring execution and debugging issues, ensuring data quality and consistency throughout the pipeline.
- **Metadata Management Excellence:** Maintained the **Data Repository Area** (DRA) to centralize metadata for **ETL** processes, improving transparency, tracking, and troubleshooting for data integration tasks.
- **Enhanced Data Quality and Stakeholder Collaboration:** Monitored and validated data in the **DSA**, resolving bottlenecks and aligning transformations with business requirements through close collaboration with cross-functional teams.
- **Schema Management Expertise:** Managed staging area schemas like **DSA_OWNER** to enforce security, optimize structure, and streamline access control, ensuring scalability and data governance.
- **Enhanced Data Quality and Performance:** Diagnosed and corrected discrepancies in staging data, refined SQL queries generated by ODI, and collaborated with database teams to improve overall ETL performance.

- Conducted thorough **data cleaning and validation** within PL/SQL to align with **business requirements** and maintain high data quality.
- **Documented ETL processes**, including design workflows, technical specifications, and user guides, for easy maintenance and team onboarding.
- Worked collaboratively in an **Agile environment**, participating in **sprint planning**, daily stand-ups, and retrospectives to ensure on-time delivery of tasks.
- Monitored and troubleshot ETL workflows and KSH executions in real time, ensuring uninterrupted data transfer and workflow performance.
- Followed coding best practices to deliver scalable, reliable, and high-quality solutions for data integration and transformation.

Environment: Oracle Database, Oracle Data Integrator (ODI), PL/SQL, KornShell (KSH), Putty, Data Staging Area (DSA), Operational Data Warehouse (ODW), Data Repository Area (DRA), Agile Methodology, UNIX/Linux.